Kareem Abdel-Hafez

Published as: K. M. Gameel | Canadian/British citizen

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Postdoctoral Fellow, University of Toronto — Clean Energy Lab; Prof. Oleksandr Voznyy
Affiliations: Acceleration Consortium (AC) & A3MD – AI for Materials Discovery
Role: Quantum Methods for Materials & Molecules | Scientific Machine Learning
Peer Reviewer: *Applied AI Letters* (Wiley), 2025

Education

- **Ph.D. in Electrical & Computer Engineering**, University of Toronto — 2024
  - Quantum Mechanical Modeling of Electrochemical Interfaces of Energy Storage Materials
  - Committee included O. A. von Lilienfeld, A. Aspuru-Guzik, & JM Garcia-Lastra.
  - GPA: 3.9/4.0; dissertation accepted without revisions
  - Hatch Graduate Scholarship for Sustainable Energy Research Award
- **M.Sc. in Physics**, American University in Cairo — 2018
  - Theoretical modeling of CO Adsorption on Metallic Surfaces from Molecular Orbital Principles
  - Award: Outstanding M.Sc. Thesis Award
- **B.Eng. in Energy & Renewable Energy Engineering**, Ain Shams University — 2014

Postdoc-Industry Collaborations

- **Merck EMD Electronics, Acceleration Consortium – Lead Scientist:** Led technical presentations and delivered software solutions integrating quantum simulations and GNNs for rapid organometallic ligand-dissociation energy prediction in semiconductor-fabrication applications.
- **Meta FAIR Chemistry Group / Ted Sargent Group, A3MD – OER Catalysis:** Linked experimental catalyst measurements with high-throughput DFT; computed electronic-structure descriptors and built neural-network models using PDOS features to predict catalytic overpotentials.
- **LG Energy, Acceleration Consortium:** Delivered DFT & ab initio molecular-dynamics modeling of degradation mechanisms in Ni-rich lithium-ion battery cathodes.

Publications (published as K. M. Gameel)

1. **[FIRST]** *"The Mechanics of Δ Learning: Target Design Approach for Transferable Molecular Models"*. In progress.
2. **[FIRST]** *"Unraveling the Quantum Capacitance of Graphene Slit Pores with Ions under Extreme Confinement"*, ACS Physical Chemistry C, 2023.
3. **[FIRST]** *"First-Principles Approach to Modeling Interfacial Capacitance in Graphene-Based Electrodes"*, ACS Physical Chemistry C, 2023.
4. **[FIRST]** *"Unveiling CO Adsorption on Cu Surfaces: New Insights from Molecular Orbital Principles"*, RSC Physical Chemistry Chemical Physics, 2018.
5. **[FIRST]** *"First Principles Descriptors of CO Chemisorption on Cu & Ni Surfaces"*, RSC Physical Chemistry Chemical Physics, 2019.
6. **[CO-FIRST]** *"The DFT+U: Approaches, Accuracy, & Applications"*, book chapter, IntechOpen, 2018.
7. **[FIRST]** *"Optimal Locations of Distributed Generations in Electrical Power Distribution System"*, 7th IEEE Conference, 2013.

8. **[THIRD]** *“Electrospun Lead-Free Double Perovskite Nanofibers for Photovoltaic and Optoelectronic Applications”*, ACS Applied Nano Materials, 2019.

Invited Talks & Conference Presentations

- *Delta Learning as a Target Design Approach for Transferable Machine Learning for Molecules & Materials*, accepted abstract for oral presentation, Canadian Societies for Chemistry & Chemical Engineering 2026 Conferences & Exhibition (CSC/CSChE x2026), Toronto, Canada, May 27, 2026.
- *Physics-Informed Machine Learning for Molecular & Materials Discovery*, Invited Talk, AUC ML Day (2025).
- *Δ-Learning for Enhanced Generalization in Molecular GNNs*, Invited Presentation, Acceleration Consortium–Seoul National University Workshop (2025).
- *Optimal Locations of Distributed Generations in Electrical Power Distribution System*, oral presentation, 7th IEEE Conference, Doha, Qatar, November 2013.

Skills

- **Computational chemistry tools:** plane-wave DFT methods including VASP and Quantum Espresso; localized-basis methods including Gaussian; mixed Gaussian/plane-wave methods including CP2K; semi-empirical quantum chemistry including GFN-xTB; classical force-field methods; ab initio and classical molecular dynamics; transition-state search.
- **Atomistic computations:** electronic-structure calculations for molecules, crystals, surfaces, and interfaces; adsorption, interaction energies, charge transfer, electrochemical interfaces, organometallic complexes, and ligand dissociation energetics.
- **Machine learning:** neural networks, graph neural networks, MLIPs/ML force fields, invariant/equivariant atomistic models, Δ-learning, target design, transferability analysis, uncertainty quantification, and baseline-to-DFT correction models.
- **HPC and software:** Python, PyTorch-style model development, automated pipelines, Slurm-based CPU/GPU cluster computing, high-throughput dataset generation, model training, evaluation, and reproducible computational workflows.

Research Experience

Postdoctoral Fellow – AI for Molecular & Materials Systems, University of Toronto 2024–Present
Department of Physical & Environmental Sciences — Supervisor: Prof. Oleksandr Voznyy

- Participate in industrial collaborations with A3MD and Acceleration Consortium partners, including Meta, LG Energy, and Merck EMD Electronics, serving as lead scientist on the Merck EMD Electronics project.
- Develop the Δ-ML project as a target-design strategy for transferable molecular models, studying how baseline choice reshapes target smoothness, variance, and feature-target mappings.
- Develop Hamiltonian-based representations for accurate bandgap prediction using graph-based neural-network models.

Ph.D. Researcher – Modelling of Energy Storage Systems, University of Toronto 2019–2024
Department of Electrical & Computer Engineering — Supervisor: Prof. Francis Dawson

- Developed first-principles models of graphene-based electrochemical interfaces, combining quantum capacitance, dielectric screening, ion-electron effects, and classical electrostatics to explain experimental capacitance trends. ***(ACS Phys. Chem. C, 2023)***

- Simulated ions under extreme nanopore confinement to resolve capacitance discrepancies & identify atomistic descriptors for charge storage. ***(ACS Phys. Chem. C, 2023)***

M.Sc. Computational Group Leader – Electronic Structure, American University in Cairo
2015–2018
Physics Department — Supervisor: Prof. Nageh Allam

- Led DFT & molecular-orbital studies of CO adsorption on transition-metal surfaces, investigating the roots of the DFT CO adsorption puzzle from molecular-orbital principles. ***(PCCP, 2018; 2019)***
- Modeled halide-perovskite electronic structure & co-authored a DFT+U methods chapter. ***(ACS, 2019; IntechOpen, 2018)***

Teaching & Mentorship

Instructor of Record – Undergraduate Foundational Physics 2018–2019
UofCanada Egypt / University of Prince Edward Island partnership

- Delivered full-term physics instruction covering mechanics, waves, thermodynamics, & problem-solving methods.
- Designed lectures, assessments, grading rubrics, & academic-support materials for undergraduate students.

Teaching Assistantships – Physics, Mathematical Physics, & Programming 2015–2024

- **University of Toronto**
  - **PHY131** – Introduction to Physics I
  - **MAT291** – Introduction to Mathematical Physics
  - **APS105** – Computer Fundamentals
- **American University in Cairo**
  - **PHYS-111** – Classical Mechanics, Sound, & Heat
  - **PHYS-112** – Electricity & Magnetism

Relevant Graduate Coursework for Teaching

- **Ph.D.:** Quantum Theory of Solids; Materials Physics; Scientific Computing for Physicists; Statistical Mechanics; Engineering Mathematics.
- **M.Sc.:** Quantum Mechanics; Advanced Quantum Mechanics; Classical Mechanics; Methods of Structural Determination in Chemistry; Mathematical Physics.

Mentorship & Teaching Development

- Led TA training workshops at AUC & mentored students in physics problem solving, computational methods, & electronic-structure research workflows.

Completed Technical Training

- **IBM Foundation Models for Materials (FM4M)** — Hands-on training & evaluation of foundation models for atomistic systems (2024).
- **AIRCHECK Workshop – Machine Learning for Drug Discovery** — Applied gradient boosting & feature optimization for molecular property prediction (2025).

Awards

- Hatch Graduate Scholarship for Sustainable Energy Research, University of Toronto (2021–2023)
- Outstanding M.Sc. Thesis Award, American University in Cairo (2019)

- Academic Honor Award, American University in Cairo (2018)
- University Graduate Fellowship, AUC (2016)